Project Proposal

Project: Investigating How Noise Characteristics Affect Speech Enhancement Performance

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1 Project Overview

Background noise can reduce the quality and intelligibility of speech recordings, particularly when the noise overlaps with speech frequencies or changes over time. Speech enhancement techniques aim to reduce unwanted noise while preserving the important parts of the speech signal.

This project will investigate how time frequency overlap and changes in background noise over time affect speech enhancement performance at the same overall signal to noise ratio (SNR). The main aim is to understand why some noisy speech recordings are more difficult to enhance than others, and whether traditional and modern enhancement methods experience similar difficulties under these conditions.

Spectral subtraction and Wiener filtering will be investigated as traditional enhancement methods, alongside a modern pretrained system such as DeepFilterNet. The project will examine whether these methods experience similar difficulties under different acoustic conditions. Their performance will be evaluated using measures of speech intelligibility and signal distortion, supported by spectral analysis.

2 Background and Motivation

Speech enhancement is used to improve speech signals that have been affected by unwanted background noise. This is important in applications such as communication systems, hearing devices and speech recognition, where environmental noise can make speech more difficult to understand.

A number of signal processing techniques have been developed for speech enhancement. Spectral subtraction is a well-established method that estimates and reduces the background noise spectrum, with early approaches developed by Boll [1] and Berouti et al. [2]. Wiener filtering is another commonly used approach that estimates the clean speech signal while reducing unwanted noise [3].

Although traditional speech enhancement algorithms have been widely studied, their performance can vary depending on the characteristics of background noise. Previous research has already compared spectral subtraction and Wiener filtering under stationary and non-stationary noise conditions [4]. Therefore, simply comparing these algorithms across different noise environments would not provide much new information.

Two speech recordings can have the same overall SNR but still be very different in how difficult they are to enhance. For example, relatively constant fan noise may overlap less with speech frequencies, while cafe noise can contain conversations that overlap strongly with speech. Traffic noise can also change significantly over time, making it more difficult to estimate and remove.

This project will investigate whether time-frequency overlap and noise non-stationarity can help explain these differences. Rather than focusing only on noise categories such as traffic or cafe noise, the project will measure the acoustic characteristics of the interference and examine how they affect the

performance of traditional and modern enhancement methods.

3 Proposed Methodology

The project will be implemented primarily using MATLAB. Clean speech recordings will be obtained from a suitable dataset such as VCTK [5], while environmental noise recordings will be selected from the DEMAND database [6]. Four different noise conditions will be considered, including white noise, relatively stationary environmental noise, traffic noise and more rapidly changing public space noise.

Noisy speech signals will be created by adding background noise to clean speech at controlled signal-to-noise ratio (SNR) levels:

$$x[n] = s[n] + \alpha n[n]$$

where $s[n]$ is the clean speech signal, $n[n]$ is the background noise and α controls the noise level. Initial testing will focus on 0 dB SNR, with additional levels of -5 dB, 5 dB and 10 dB considered in later experiments.

Short-Time Fourier Transform (STFT) analysis will be used to examine how speech and noise change across time and frequency. Two acoustic properties will be measured: time-frequency overlap and noise non-stationarity.

Time-frequency overlap measures how much speech and noise energy occupy the same time-frequency regions:

$$O = \frac{\sum_{k,m} \min(|S(k,m)|^2, |N(k,m)|^2)}{\sum_{k,m} |S(k,m)|^2 + \epsilon}$$

where $S(k,m)$ and $N(k,m)$ represent the STFTs of clean speech and the scaled noise present in the mixture. Larger values of O indicate greater overlap between speech and noise.

Noise non-stationarity measures how much the noise spectrum changes between consecutive time frames. The normalised noise spectrum is defined as:

$$p_N(k,m) = \frac{|N(k,m)|^2}{\sum_j |N(j,m)|^2 + \epsilon}$$

The spectral variation is then calculated as:

$$V_N = \frac{1}{M-1} \sum_{m=2}^{M} \|p_N(:,m) - p_N(:,m-1)\|_2$$

Larger values indicate that the frequency distribution of the noise changes more rapidly over time.

An Ideal Ratio Mask (IRM) will also be examined to identify speech-dominant and noise-dominant time-frequency regions:

$$M_{\text{IRM}}(k,m) = \frac{|S(k,m)|^2}{|S(k,m)|^2 + |N(k,m)|^2 + \epsilon}$$

The IRM uses the known clean speech and noise signals and will be used as an oracle reference rather than a practical enhancement algorithm. Ideal ratio masking has also been investigated in previous speech separation research [7].

Spectral subtraction and Wiener filtering will be implemented in MATLAB using STFT analysis, noise estimation and inverse STFT reconstruction. Existing implementations will be examined to help verify the results. A pretrained model such as DeepFilterNet [8] will also be considered as a modern enhancement reference, without training a neural network.

The enhanced speech will be compared with the original clean speech using Short-Time Objective Intelligibility (STOI) [9] and Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) [10]. Spectrograms

and listening comparisons will also be used to examine speech distortion and remaining background noise.

The results will be analysed to determine whether time-frequency overlap and noise non-stationarity help explain differences in enhancement performance, particularly when different noisy recordings have the same overall SNR.

4 Expected Outcomes

The project is expected to produce a MATLAB-based investigation of how the characteristics of background noise affect speech enhancement performance under controlled SNR conditions.

The main aim is to determine whether time-frequency overlap and noise non-stationarity can help explain why some noisy speech recordings are more difficult to enhance than others, even when their overall SNR is the same.

Spectral subtraction and Wiener filtering will be evaluated as traditional enhancement methods, with a pretrained modern system included where feasible. The results will be assessed using objective measures such as STOI and SI-SDR, supported by spectrograms and listening comparisons.

The investigation will aim to identify conditions where enhancement performance becomes limited, such as when noise strongly overlaps with speech or changes rapidly over time. The results will be used to examine whether traditional and modern enhancement methods experience similar limitations.

The MATLAB implementations, results, figures and project documentation will be maintained through the GitHub repository so that the experiments can be reproduced.

5 Project Timeline

Table 1: Proposed project timeline.

Weeks	**Planned Work**
1–3	Select the project topic and develop the initial research question.
4–5	Review relevant literature, incorporate supervisor feedback and refine the research question.
6–7	Select suitable speech and noise datasets, create controlled SNR mixtures and implement initial STFT analysis.
8–9	Calculate time-frequency overlap and noise non-stationarity measures, and implement spectral subtraction and Wiener filtering.
10–11	Evaluate enhancement performance using STOI and SI-SDR, and compare the results with a pretrained modern enhancement system.
12–13	Analyse relationships between acoustic characteristics and enhancement performance, investigate failure cases, complete the final report and finalise GitHub documentation.

A Preliminary Results

An initial experiment was conducted using one clean speech recording mixed with four different noise conditions at the same overall SNR of 0 dB. The noise conditions included white noise, kitchen noise, traffic noise and public-space noise.

Time-frequency overlap (O) and noise spectral variation (V_N) were calculated using STFT analysis. Spectral subtraction and Wiener filtering were then applied to each mixture. Enhancement performance was evaluated using SI-SDR improvement relative to the original noisy speech.

Table 2: Preliminary results at 0 dB input SNR. SI-SDR improvements are measured relative to the original noisy speech.

Noise	O	V_N	**SS (dB)**	**Wiener (dB)**
White	0.039	0.047	4.5	6.5
Kitchen	0.027	0.176	4.1	6.0
Traffic	0.125	0.274	2.0	2.6
Public space	0.167	0.239	1.6	2.1

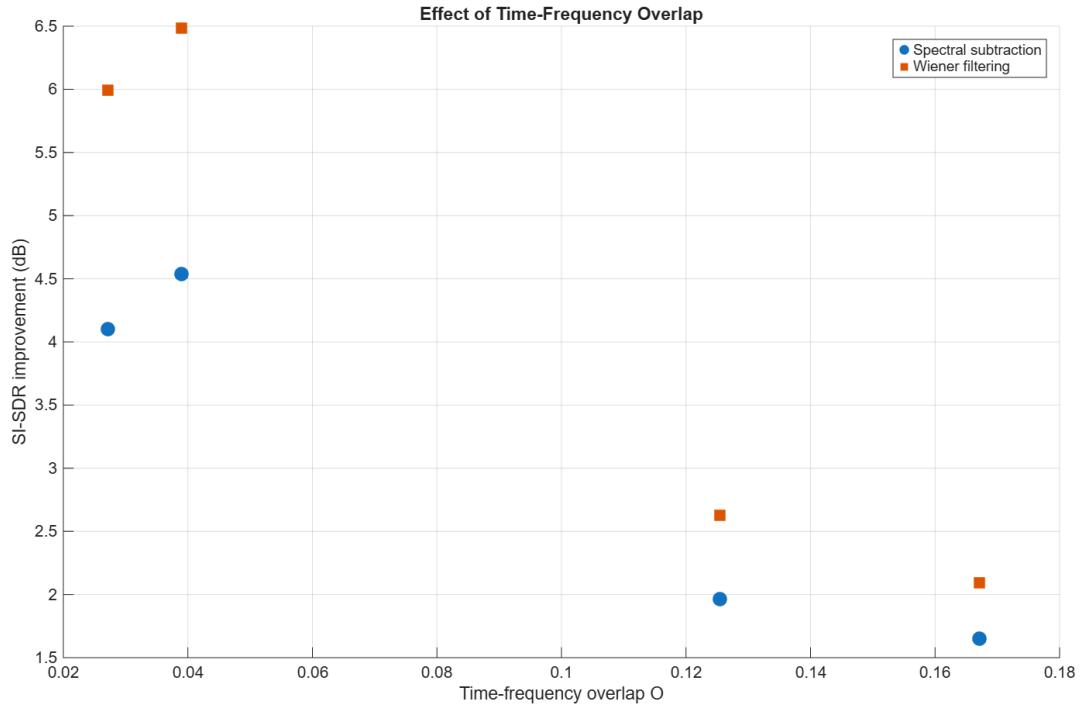


Figure 1: Relationship between time-frequency overlap and SI-SDR improvement for spectral subtraction and Wiener filtering at 0 dB input SNR.

Figure 1 suggests that greater time-frequency overlap is generally associated with smaller SI-SDR improvements. Traffic and public-space noise had higher overlap values and produced smaller improvements than white and kitchen noise. This suggests that the amount of overlap between speech and noise may help explain differences in enhancement performance, even when the overall SNR is the same.

However, the relationship was not consistent across all four conditions, indicating that other acoustic properties may also influence enhancement performance.

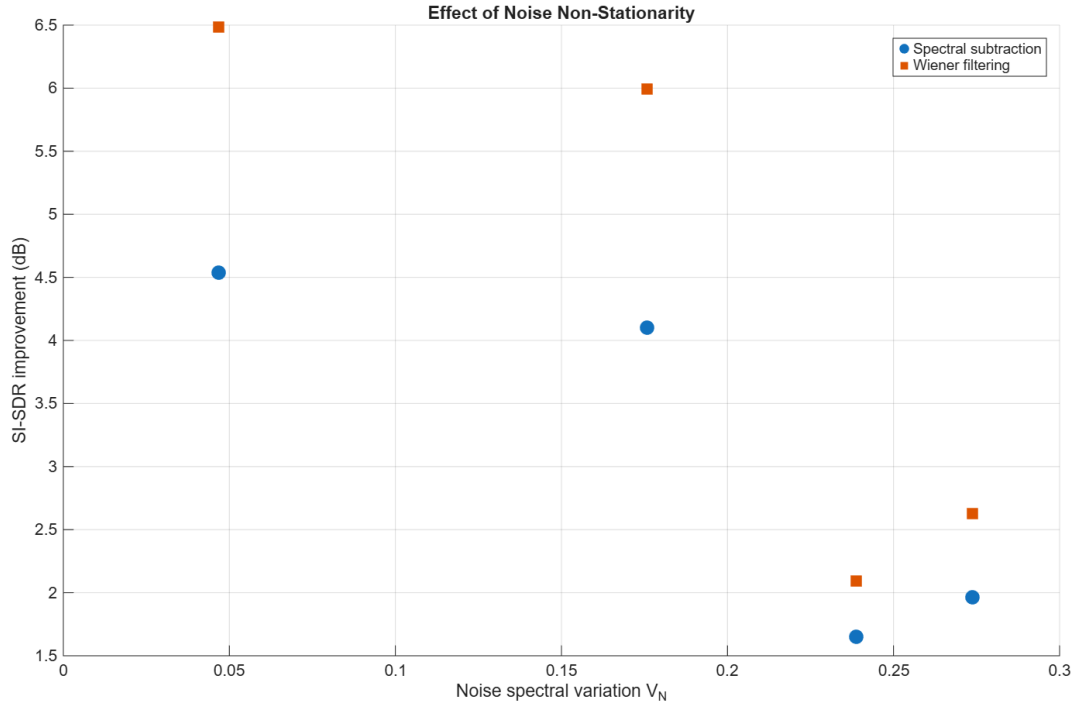


Figure 2: Relationship between noise spectral variation and SI-SDR improvement for spectral subtraction and Wiener filtering at 0 dB input SNR.

Figure 2 shows that noise spectral variation also differed between conditions, although its relationship with enhancement performance was less consistent. Traffic noise had the highest measured spectral variation, while public-space noise produced the smallest enhancement improvements.

This suggests that noise non-stationarity alone may not fully explain enhancement difficulty. Both time-frequency overlap and spectral variation may need to be considered when investigating why some noise conditions are more difficult to enhance.

Overall, both spectral subtraction and Wiener filtering improved SI-SDR for the four tested mixtures. Wiener filtering produced larger improvements in this initial experiment, although comparing the two algorithms is not the main objective of the project.

These findings are preliminary because only one clean speech recording was used. The enhancement implementations also require validation against established reference methods. Further testing with additional speech recordings and STOI evaluation will be needed before drawing broader conclusions about the relationship between acoustic characteristics and speech enhancement performance.

References

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